The Rider's Co-Pilot: An Intelligent Assistant for Smarter Riding

Capstone Project Report

Submitted in partial fulfillment for the award of the degree of

Bachelor of Technology
In

ELECTRONICS AND COMMUNICATION ENGINEERING

(AI AND CYBERNETICS)

Submitted to

VIT BHOPAL UNIVERSITY (M. P.)

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April 2026

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DECLARATION

We hereby declare that the Dissertation entitled ***"The Rider's Co-Pilot: An Intelligent Assistant for Smarter Riding"*** is our work conducted under the supervision of **Dr. Soumitra K. Nayak**, Assistant Professor, School of Electrical and Electronics Engineering (SEEE), VIT University, Bhopal.

We further declare that to the best of our knowledge, this report does not contain any part of work submitted for the award of any degree in this university or any other university / Deemed University without proper citation.

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CERTIFICATE

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Acknowledgment

First, we would like to express our gratitude to “Dr. Soumitra K. Nayak” for his supervision, advice, and guidance from the very early stage of this thesis work and for providing us with extraordinary experiences throughout the work. Above all and the most needed, he offered us unflinching encouragement and support in various ways. His words have always inspired us to work efficiently and comprehensively. We would like to thank him for his constant encouragement, which enabled us to grow as people. His presence has definitely improved us as human beings.

We express our gratitude towards “**Dr. Suresh M.”,** Associate Professor and Dean, School of Electrical and Electronics Engineering Department, VIT Bhopal University, for providing us with all the help and permitting us to work in the laboratory with no time limits.

We shall ever remain indebted to “**Prof. (Dr.) Satish Kumar Modh,”** Vice Chancellor, VIT Bhopal University, for providing us with institutional and administrative facilities during my project work at VIT.

We are very thankful to all who have supported us in completing this project.

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Abstract

The Rider's Co-Pilot is an advanced safety system built into motorcycles to enhance rider safety. It constantly checks for important safety rules before and during riding. The system helps prevent accidents caused by careless behavior, such as not wearing a helmet, riding under the influence of alcohol, or being too tired to ride properly. It combines several safety tools such as detecting if a helmet is on, checking for drowsiness, sensing alcohol levels, and preventing theft, all working together in one smart setup. In the second part of the project, the team made significant improvements to how the system communicates and perceives things.

They changed the old wireless module, which used radio frequency, to one that uses LoRa, which sends data more reliably and over longer distances. While setting this up, they faced issues such as a poor antenna signal and weak signals, but they found ways to fix these problems and improve communication. The system also uses a Raspberry Pi to detect signs of drowsiness in real time via computer vision. It looks at how wide the eyes are open (Eye Aspect Ratio) and how far the mouth is open (Mouth Aspect Ratio) to tell if someone is getting sleepy or yawning. To make the system more accurate, they added noise filters to block distractions and ensured it could still work even if only one eye was working. By combining all these features, the system keeps a close eye on the rider and makes smart decisions.

If it detects something unsafe, such as the rider not wearing a helmet or being too tired, it either stops the engine or warns them. This helps keep riders safe, lowers the chance of accidents, and encourages better riding habits.

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List of Symbols & Abbreviations

2D	Two-dimensional
EAR	Eye Aspect Ratio
MAR	Mouth Aspect Ratio
IR Sensor	Infrared Sensor
MQ-3	Gas Sensor used for alcohol detection
IEEE	Institute of Electrical and Electronics Engineers
LoRa	Long Range Wireless Communication Technology
GPIO	General Purpose Input Output
OpenCV	Open-Source Computer Vision Library
Dlib	Machine Learning Toolkit for facial landmark detection
FPS	Frames Per Second

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Chapter 1

INTRODUCTION

Road safety remains a major global problem, especially in places where motorcycles are a popular mode of transportation. Many road accidents occur not because of broken vehicles but because of mistakes people make, such as not wearing a helmet, riding while drunk, or driving while tired. These actions greatly raise the risk of serious harm or even death. Because of this, it's important to create systems that can help riders follow safety rules more effectively. As technology improves, especially in smart and embedded systems, there's a greater chance of building solutions that don't just remind people about safety but actually help them follow the rules. The Rider's Co-Pilot was created with this goal in mind. Instead of relying solely on the rider to make the right choices, this system uses an automated process that continuously checks for safety issues and takes action when unsafe conditions are detected. This helps reduce the need for human self-control and improves overall safety.

The system has two main parts that work together: a helmet unit and a vehicle unit. The helmet unit gathers real-time information about the rider, such as whether they're wearing a helmet or seem tired. This data is then sent wirelessly to the vehicle unit, which reviews it and decides whether to start or keep the motorcycle running. This ongoing exchange between the two parts is the key to how the system works. In the second phase of development, the system was improved to work more reliably. The communication system was updated by switching from an old RF module to LoRa technology, which offers a longer range and more stable connection. Also, a Raspberry Pi-based vision system was added to detect drowsiness using facial recognition techniques. These upgrades make the system more dependable, efficient, and ready for real-world use, helping to make riding safer and smarter.

1.1. Motivation

The idea for "The Rider's Co-Pilot" began with growing concerns about road safety and the rising number of motorcycle accidents. Motorcycles are popular because they are inexpensive, easy to use, and work well in traffic. But this ease of use often comes with greater risk, especially when people don't follow safety rules. Many accidents aren't because of broken parts but because of bad behaviour, like not wearing a helmet, riding drunk, or being tired or distracted. These actions can be avoided, but they still lead to serious injuries and even deaths. A big challenge is that safety rules depend mostly on people being responsible. Even though there are traffic laws, many riders don't follow them. This can be because they don't know the rules, are too confident, or there's not enough strict enforcement. In many places, especially in developing countries, there aren't effective systems to track riders who follow the rules. This shows a strong need for a system that doesn't just give advice but actively enforces safety in a reliable, automated way. Another reason for this project is the rapid growth of technology in embedded systems and smart automation. With affordable microcontrollers, sensors, and communication tools, it's now possible to build systems that monitor real-time conditions and make decisions without human involvement. This means moving from just giving warnings to using active systems that can control the motorcycle based on safety settings. Besides safety, vehicle theft is also a big issue. Traditional security systems are often easy to trick, so they don't work well in real life. By connecting the rider's helmet and their condition to the motorcycle's ignition, an extra layer of security can be added. This means the bike can't be used without proper permission, making it harder for people to steal the vehicle.

This project also aims to create a simple, widely usable solution. Instead of building a complicated and costly system, the goal is to make a low-cost model that can be used on many different motorcycles. This makes the system more affordable and easier to use, especially in places where money is tight. In the end, the main goal is to reduce accidents and promote safer riding habits. By using technology to enforce safety, the system tries to create a smarter and more secure way to ride. This helps towards the bigger goal of building smarter transportation systems.

1.2. Objective

Helmet Usage Enforcement: The system is designed to ensure the motorcycle starts only when the rider is properly wearing a helmet. The main goal is to stop people from riding without a helmet, which is a common mistake. The system uses sensors inside the helmet to check if it's being worn. If the helmet isn't detected, the motorcycle won't start, helping to keep the rider safe.

Prevention of Drunk Driving: Another important goal is to cut down on accidents that happen because of drinking alcohol. The system has a sensor that continuously monitors the rider's breath. If the alcohol level goes above a set limit, the system stops the engine from starting, making sure the rider can't ride the vehicle when they're too drunk. This is a safety feature that prevents problems before they happen, rather than just letting the rider know they're in trouble.

Drowsiness Detection and Fatigue Monitoring: The project also aims to minimize accidents caused by rider fatigue. Using vision-based techniques, the system monitors eye movements and facial behaviour to detect signs of drowsiness. If fatigue is detected, the system can either alert the rider or restrict the ignition, thereby preventing unsafe riding conditions and enhancing overall road safety.

Anti-Theft Protection Integration: In addition to safety features, the system also includes a security measure to deter theft. It connects the helmet's recognition with the motorcycle's starting system. The bike can only be turned on when the right helmet is worn, so only someone with the correct helmet can ride it.

Reliable Embedded System Design: The project involves creating a robust embedded system that uses microcontrollers such as Arduino and wireless communication modules. It makes sure there's smooth communication between the helmet part and the vehicle part, allowing for real-time monitoring and better decisions to enhance performance and reliability.

Cost-Effective, Scalable Solution and Road Safety Awareness: Another key goal of this project is to create a solution that is both low-cost and flexible enough to work with various types of two-wheelers, enabling widespread use in real-world situations. The system is built to be simple and cost-effective, so it can be easily added without needing complicated changes. Also, the project wants to improve road safety by automatically performing important safety checks that riders usually skip.

Chapter 2

LITERATURE REVIEW

This study focuses on developing a smart vehicle safety system that leverages embedded technology, computer vision, and wireless communication. The system includes features such as detecting whether a helmet is worn, identifying signs of drowsiness using EAR and MAR metrics, and detecting alcohol presence with an MQ-3 sensor. It also uses LoRa technology to enable real-time monitoring and improve rider safety.

In “Smart Helmet for Accident Detection and Alert System” Chandran, S., Chandrasekar, S. and Elizabeth [4] represent road accidents claim the lives of a significant number of individuals every day worldwide, particularly when two-wheelers are involved. This can be prevented by providing emergency care to the victim, wearing helmets, and not drinking before driving. The Smart Helmet is used as an efficient solution to the issue. The Smart Helmet serves as a key in this situation. The sensor will detect the driver's alcohol consumption and lock the engine immediately. Using GPS and GSM technologies, smart helmets offer assistance in the event of an accident. In the event of an accident, a message is sent to the ambulance or a family member with the location so that medical assistance can be given to the individual as quickly as possible. The project's goals include intelligence security, helmet wearing awareness, and human life safety prevention. Keywords- GSM, GPS, Accident.

In " Smart Helmet Using GSM & GPS Technology for Accident Detection and Reporting System," Sudarshan, Raj & Shetty, Manjesh N. [5] present a wide Smart Helmet Using GSM & GPS Technology for Accident Detection and Reporting System. A smart helmet is a special idea which makes motorcycle driving safer than before. This is implemented using GSM and GPS technology. The working of this smart helmet is very simple; vibration sensors are placed in different places of helmet where the probability of hitting is more which are connected to microcontroller board. So when the rider crashes and the helmet hit the ground, these sensors sense and gives to the microcontroller board, then controller extract GPS data using the GPS module that is interfaced to it.

In “Smart Safety Helmet for Bike Riders using IoT” Dharani, P., Ganesh, T., Gopinath, V. and Vali [6] according to the law of the Indian government as per section 129 of the motor vehicle act of 1988 briefly explains the motorcycle or two-wheeler rider is mandatory to wear the helmet while driving and the recent survey conducted on road accidents by the world health organization (WHO). This WHO organisation has briefly discussed the causes and prevention of road accidents worldwide. They also mentioned the highest death rate that took place in India and the survey also reported as per the rate 1.5 lakh of road death has been accounted for by each year approximately. The above article motivates us to develop a system that is capable of providing safety and precaution to the bike rider. We designed a system that is capable of detecting the rider is whether wearing the helmet or not. Then detecting if the rider has consumed alcohol or not, whether if these two conditions are yet satisfied then only the motor will ignite or else it will not ignite. In the event of an accident, our system can detect it and estimate its location approximately. We installed the LED strip indicator in the helmet unit to reduce the number of accidents at night.

In the work " Smart Helmet. IOP Conference Series” Sobhana, S & Sowmeeya, S & Srinathji, M & Tamilselvan [7] according to the investigation in India, nearly 25% of the road accidents are caused by two wheelers. The foremost causes for the fatalities are due to drunken driving, rash driving, and drowsiness due to long drives. The aim is to build an interesting smart helmet that protects us from accidents and indicates the accident-prone area. Here we are using various sensors to build the smart helmet. To detect alcohol consumption of the rider we use alcohol sensors. In order to check a rider’s helmet, an infrared sensor can be used. A vibration detector is also added to the helmet to indicate the harsh hitting of the helmet during an accident. When the two wheelers slide down due to road rashes, the GPS is used to identify the location of the accident spot and quickly sends messages (location) to police stations and hospitals nearby through GSM.

In “Smart helmet to start the motorbike and to prevent accidents “J Joy Mathavan [8] most of the motorbike riders meet a fatal end due to severe head injuries during accidents. Usually in Asian countries, riders are reluctant to wear helmets while riding motorbikes due to carelessness and negligence. This research aims at reducing the accidents that may occur due to such carelessness and at the same time, immediately report the accident to the emergency contact numbers. This research work is aimed at

making the necessary conditions essential to start the motorbike. The rider needs to wear the helmet, should be free from consumption of alcohol and should take off the side stand before igniting the vehicle; failure of any one of this condition would not ignite the engine. The proposed system detects the accident occurrence as well. SMS will be sent to the emergency numbers through the GSM module. The prototype works on start mode, running mode and accident mode with different functionalities in each mode. The rider needs to clear the start on mode to ignite the engine and a smart system will continuously monitor the bike and rider in the running mode. In case of unexpected interruptions during running mode, the proposed system enters into accident mode and enables GSM to report the accident to the pre-registered phone number.

In “Driver Drowsiness Detection and Alert System” Titare, Swapnil & Chinch hare, Shubham & Hande, K [9] nowadays, accidents occur during drowsy road trips and increase day by day; It is a known fact that many accidents occur due to driver fatigue and sometimes inattention, this research is primarily devoted to maximizing efforts to identify drowsiness. State of the driver under real driving conditions. The aim of driver drowsiness detection systems is to try to reduce these traffic accidents. The secondary data collected focuses on previous research on systems for detecting drowsiness and several methods have been used to detect drowsiness or inattentive driving. Our goal is to provide an interface where the program can automatically detect the driver's drowsiness and detect it in the event of an accident by using the image of a person captured by the webcam and examining how this information can be used to improve driving safety can be used. a vehicle safety project that helps prevent accidents caused by the driver's sleep. Basically, you're collecting a human image from the webcam and exploring how that information could be used to improve driving safety. Collect images from the live webcam stream and apply machine learning algorithm to the image and recognize the drowsy driver or not. When the driver is sleepy, it plays the buzzer alarm and increases the buzzer sound. If the driver doesn't wake up, they'll send a text message and email to their family members about their situation. Hence, this utility goes beyond the problem of detecting drowsiness while driving. Eye extraction, face extraction with dlib.

2.1 Identified Challenges and Strategic Points

2.1.1 Communication Latency:

In smart safety systems [8], maintaining reliable communication between different modules is a major challenge. Since the system uses wireless communication through LoRa, signal loss or delay can affect the coordination between the helmet unit and the vehicle system. Any interruption in communication may lead to incorrect decision-making, such as failure to control ignition.

To overcome this, LoRa technology is used due to its long-range and low-power capabilities. It ensures stable and continuous data transmission between modules. This improves system reliability and enables real-time response, even over larger distances.

To address this challenge, LoRa technology is used because it offers long-distance communication and uses very little power. LoRa is built to maintain reliable data transfer even in places where regular wireless systems might not work well. It sends data over long distances with very little loss and keeps the connection stable between different parts of the system. The system also keeps sending and receiving signals all the time, which helps prevent communication issues. This makes the whole system more dependable and allows for real-time responses, making sure that important safety functions like controlling ignition and sending alerts work correctly and on time.

2.1.2 Dynamic Environmental Changes:

In real driving situations, the system has to work in environments that are always changing, which makes it hard to keep performing reliably. Things like different lighting, moving backgrounds, camera noise, and where the rider is sitting can all affect how well the detection system works. For example, when light levels change between day and night, or when there are shadows or reflections, it can mess up the system's ability to detect facial features. This can cause the EAR and MAR values to fluctuate. Also, things like moving objects in the background or having multiple faces in the frame can confuse the system and lower its accuracy. These constantly changing conditions mean the system needs to be very flexible and able to keep working well no matter what. If these changes aren't handled properly, it might lead to wrong alerts about drowsiness or false alarms, which can make users lose trust in the system. So, making

sure the system works the same way in all kinds of environments is really important for safety features that operate in real time. To deal with this, the system uses adaptive methods like noise filtering and focusing on the face. The algorithm looks only at the area where the face is and ignores other background details. This helps keep facial feature detection steady and makes EAR and MAR calculations more accurate. Also, using thresholds to make decisions helps reduce false alarms by looking at several frames in a row instead of just one.

Additionally, the system can handle different ways people sit and move. It keeps track of facial features and adapts to small movements, making sure the monitoring doesn't stop. These adaptive features let the system work well even in tough and unpredictable situations. As a result, the system becomes more reliable, accurate, and better at real-time performance, making it ready for use in smart vehicle safety systems.

Chapter 3

PROPOSED METHODOLOGY

3.1. Selected Algorithms

3.1.1. Eye Aspect Ratio (EAR) Based Drowsiness Detection Algorithm

The Eye Aspect Ratio (EAR) algorithm is used to detect when someone is sleepy by looking at where the eyes are and how they move. It works by finding a ratio that compares the distance between points on the top and bottom of the eye to the distance between points on the left and right sides. This ratio helps figure out if the eyes are open or shut. Here are the steps that explain how the algorithm works.

1. **Initialization:** The system begins by turning on the camera and bringing in the necessary libraries, like OpenCV and Dlib. It also loads a pre-trained model that helps find the points on the face, especially around the eyes. The EAR threshold is set to 0.25, and a counter for the frames is also started.

2. **Frame Capture:** The system continuously captures video frames from the camera in real time for processing.

3. **Face and Landmark Detection:** For every frame, the system finds the rider's face and uses the Dlib facial landmark detector to get the positions of the eye points.

4. **EAR Calculation:** The Eye Aspect Ratio is determined by measuring the distances between the vertical and horizontal points on the eyes. This number shows the current state of the eye.

5. **Condition Checking:** The calculated EAR value is compared with the threshold value of 0.25. If the EAR is greater than the threshold, the eyes are considered open.

6. **Frame Counter Update:** If the EAR falls below 0.25, the system increases a counter to track how long the eyes remain closed.

7. **Drowsiness Detection:** If the EAR remains below the threshold for more than 20 consecutive frames, the system identifies the rider as drowsy.

8. **Alert Generation:** When drowsiness is detected, the system displays a warning message such as “DROWSY!” and can trigger an alarm. If the eyes reopen, the counter resets and the system continue monitoring.

9. **Continuous Monitoring:** The entire process runs continuously in real time to ensure constant monitoring of the rider’s condition.

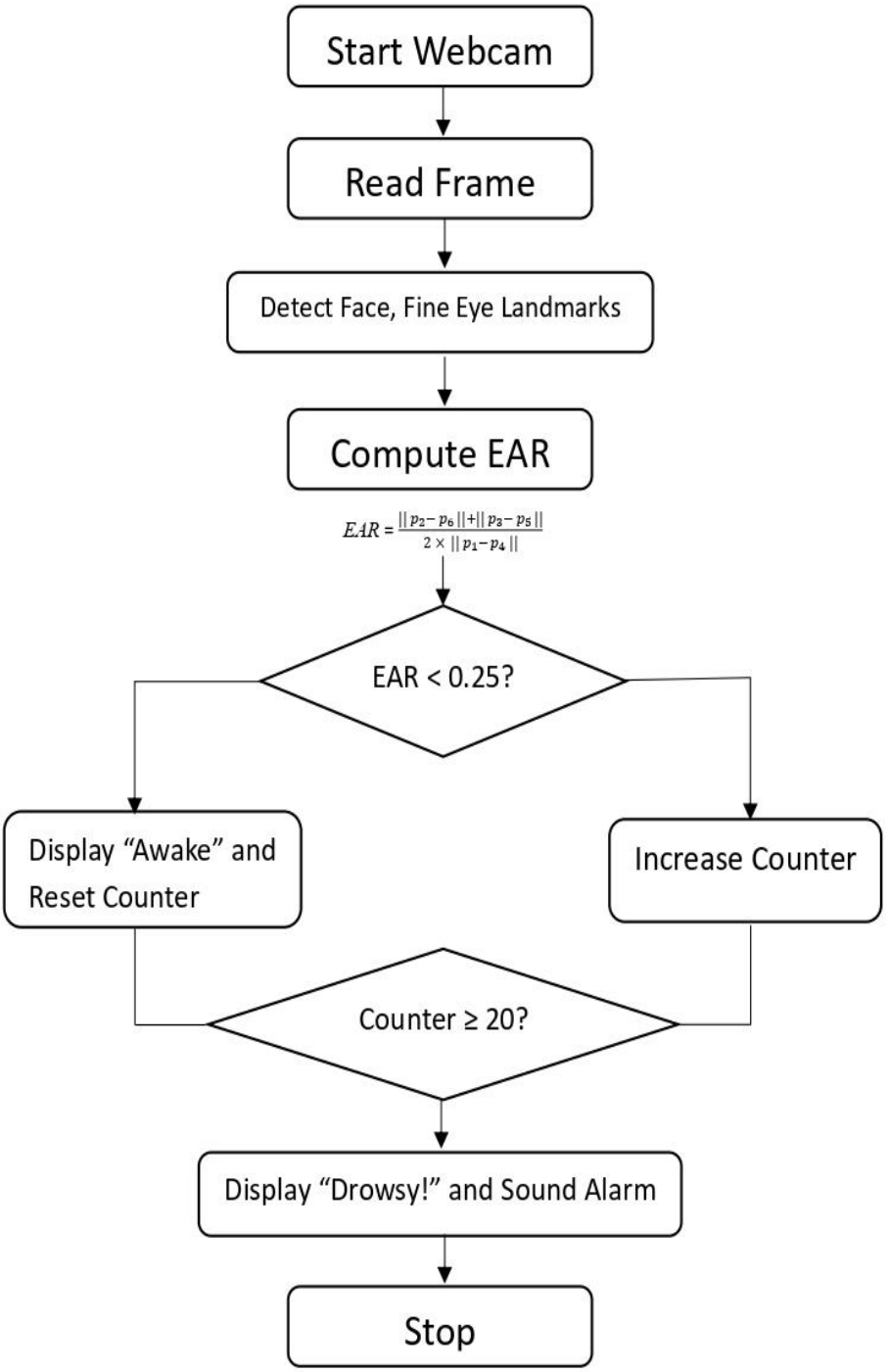


Fig 3.1 Flowchart for EAR Algorithm

Rationale for Selection: The Eye Aspect Ratio (EAR) approach is chosen for this project because it is simple, efficient, and works well for real-time drowsiness detection. Unlike more complicated deep learning methods that need a lot of data, powerful computers, and long training times, the EAR method uses basic geometry based on facial landmarks. This makes it very efficient and easy to use on systems like a Raspberry Pi, which have limited computing power. One big advantage of this method is how well it can tell the difference between open and closed eyes.

When the eyes are open, the EAR value stays pretty much the same, but it drops a lot when the eyes are closed. This clear change lets the system detect drowsiness accurately without needing complex setup or extra processing. Using a set threshold, like 0.25, makes it easy to decide if someone is tired while still being effective. Another reason this method is good is that it works continuously in real time.

The algorithm checks each video frame on its own, so the system can keep track of eye activity the whole time. This constant monitoring helps catch signs of tiredness right away, which makes the system safer for riders. Also, the EAR method is very flexible and can be improved with other techniques like noise filtering to cut through background distractions or partial eye detection for people with only one working eye. These additions make the system more reliable in different conditions, such as poor lighting or when there is camera noise.

In summary, the EAR-based method is a great choice because it is low cost, easy to implement, works in real time, and performs reliably.

It’s perfect for building an effective and practical drowsiness detection system.

3.1.2. Mouth Aspect Ratio (MAR) Based Yawning Detection Algorithm

The Mouth Aspect Ratio (MAR) algorithm detects yawning by looking at how the mouth moves and opens using facial landmark points. It tracks how wide the mouth opens over time and checks if the rider is yawning. This method works because the mouth opens a lot during a yawn, which makes the MAR value go up. The algorithm uses facial landmark detection with Dlib and OpenCV to process this in real time.

1. **Initialization:** The system starts by turning on the camera and bringing in necessary tools like OpenCV and Dlib. It then loads a tool that helps find important points around the mouth. Also, it sets up some limits.
2. **Frame Capture:** The system continuously captures frames from the camera in

real time to monitor the rider's facial movements.

3. **Face and Landmark Detection:** For each frame, the system detects the face and extracts mouth landmark points (p60 to p67) using the Dlib facial landmark detector.
4. **MAR Calculation:** The mouth aspect ratio is determined by measuring the distance between vertical and horizontal landmarks on the mouth. The formula compares the vertical opening of the mouth to its horizontal width, giving a numerical value that shows how open the mouth is.
5. **Condition Checking:** The MAR value is checked against a set limit. If the mouth is shut, the MAR stays at a low level. But when the mouth opens wide, like during a yawn, the MAR goes up a lot.
6. **Alert Generation:** If the system notices someone yawning, it can send a warning or alert to show they might be tired. This can work together with detecting drowsiness to improve safety.
7. **Continuous Monitoring:** The algorithm works all the time in real time, keeping an eye on the rider's face during the whole trip.

8. **Frame Capture:** The system continuously captures frames from the camera in real time to monitor the rider's facial movements.
9. **Face and Landmark Detection:** For each frame, the system detects the face and extracts mouth landmark points (p60 to p67) using the Dlib facial landmark detector.
10. **MAR Calculation:** The mouth aspect ratio is determined by measuring the distance between vertical and horizontal landmarks on the mouth. The formula compares the vertical opening of the mouth to its horizontal width, giving a numerical value that shows how open the mouth is.
11. **Condition Checking:** The MAR value is checked against a set limit. If the mouth is shut, the MAR stays at a low level. But when the mouth opens wide, like during a yawn, the MAR goes up a lot.
12. **Alert Generation:** If the system notices someone yawning, it can send a warning or alert to show they might be tired. This can work together with detecting drowsiness to improve safety.
13. **Continuous Monitoring:** The algorithm works all the time in real time, keeping an eye on the rider's face during the whole trip and quickly spotting any signs of tiredness.

Rationale for Selection: The Mouth Aspect Ratio (MAR) based approach is chosen for this project because it offers a reliable and effective way to detect yawning, which is a key sign of fatigue and drowsiness. Unlike methods that rely solely on visual observation, the MAR approach uses precise geometric relationships between specific facial landmark points to measure how open the mouth is. This allows the system to reliably differentiate between everyday mouth movements and true yawning, which significantly improves the accuracy of detection. One of the main benefits of the MAR method is its simplicity and ease of use. It doesn't need large datasets, complex training, or high computational power, making it well-suited for real-time applications. The algorithm works directly with facial landmarks detected using the Dlib library and processes each frame efficiently using OpenCV. This makes it highly compatible with resource-constrained environments such as Raspberry Pi, where processing power and memory are limited.

Another key reason for selecting this approach is its ability to minimize false positives.

In real-world settings, people may open their mouths briefly during speech or other facial expressions. The MAR method tackles this by tracking the ratio over several consecutive frames instead of relying on just a single frame. A yawn is only detected when the MAR value remains above a threshold for a specific period, which enhances the system's reliability and accuracy. Additionally, the MAR algorithm works well with the EAR-based eye detection system.

While EAR is focused on eye closure, MAR captures yawning behavior, offering a more complete picture of a rider's fatigue state. By combining both methods, the system achieves better accuracy and overall performance.

In summary, the MAR-based method is ideal for yawning detection in intelligent rider safety systems due to its efficiency, accuracy, real-time capability, and compatibility with embedded platforms.

Implementation Strategy: The strategy for detecting yawning based on Mouth Aspect Ratio (MAR) involves analyzing how the mouth moves and opens using facial landmark detection. The system starts by capturing live video through a camera, which is then processed using Python, along with OpenCV and Dlib. For each frame, facial landmarks are detected, with particular attention to the mouth area. Specific key points, from p60 to p67, are used to calculate the MAR value. The MAR is determined by comparing the vertical distance between the top and bottom of the mouth to the

horizontal width of the mouth. This ratio gives a numerical indication of how wide the mouth is open. The system keeps checking the MAR value for every frame and compares it to a set threshold. If the MAR value goes above this threshold for several consecutive frames, the system determines that a yawn is happening. This method helps avoid misclassifying brief or regular mouth movements as yawns.

To make the detection more accurate, the system uses frame-based checks, requiring the condition to last across multiple frames before confirming a yawn. This helps prevent false alarms caused by talking or slight facial expressions. The system is designed for real-time use and can run smoothly on devices like a Raspberry Pi, allowing for ongoing monitoring without delays.

Operational Benefits: The MAR-based method for detecting yawning has several benefits that help make riding safer. One key advantage is its ability to spot tiredness by looking at yawning, which is a clear sign of someone getting sleepy. Unlike systems that depend on just one thing, MAR adds another way to detect this, which works well with eye-based methods, leading to more accurate results. This method is also fast to run and doesn’t need complicated setup or a lot of data, making it perfect for use in real time. It works reliably in different situations because it focuses on shapes and how things move rather than colors or patterns. Looking at several frames in a row also helps the system be more accurate by reducing false alarms.

Another benefit is that combining MAR with other tools like EAR lets the system understand the rider's condition better. This approach of using multiple signals makes the detection more dependable and helps give timely warnings, which helps keep riders safer.

Strategic Integration: The MAR-based yawning detection module is built to work smoothly with other safety features in the system. It pairs with the EAR-based drowsiness detection module to give a full picture of the rider's fatigue level. By using both eye and mouth cues, the system becomes more accurate and dependable in identifying unsafe situations.

Additionally, the MAR module can connect with the main decision-making system that handles vehicle ignition and alerts. If yawning is detected along with other signs of tiredness, the system can send warnings or take steps to prevent accidents. This connection helps the system react quickly and effectively to changing conditions in real time. In general, the MAR algorithm is important for making the system work better by adding another reliable way to measure fatigue. This improves safety and makes the system more effective in real-life use.

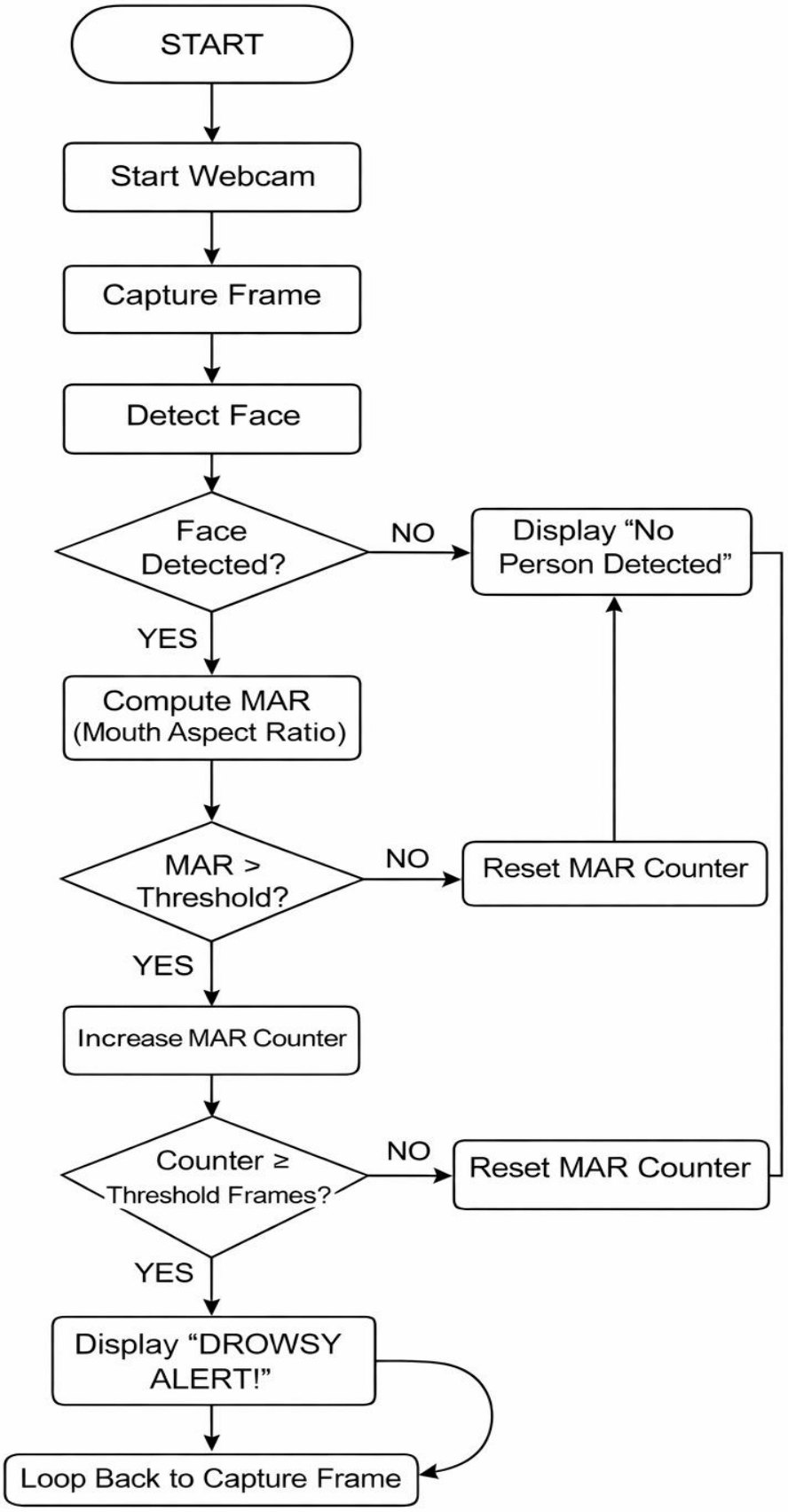


Fig 3.2 Flowchart for MAR Algorithm

3.2. Decision Function

To make sure the motorcycle operates safely and lower the risk of accidents due to unsafe rider behavior, a decision system is created by combining several safety factors. The system checks different things like whether the rider is wearing a helmet, their eye condition using Eye Aspect Ratio (EAR), their yawning behavior using Mouth Aspect Ratio (MAR), and if they have alcohol in their system. Each of these factors plays a part in the final decision, which decides if the motorcycle's ignition should be turned on or kept off.

Helmet Detection: This part checks if the rider is wearing the helmet correctly. It is the main safety check that needs to be satisfied before the system can start working. The result from this part is shown like this:

$H(x) = 1$, when the helmet is worn

$H(x) = 0$, if the helmet is not worn

If the helmet isn't detected, the ignition system stays turned off no matter what other conditions are met.

Drowsiness Detection (EAR): This part checks how alert the rider is by looking at how their eyes are closed using something called the Eye Aspect Ratio. The EAR is calculated like this:

$$EAR = (p2 - p6 + p3 - p5) / (2 \times p1 - p4) \quad (3.2.1)$$

When the eyes are open, the EAR stays pretty high, around 0.3.

But when the eyes are closed, it goes below a certain number, which is 0.25. If the EAR is less than 0.25 for 20 straight frames, the rider is thought to be sleepy:

$$E(x) = 0\ (sleepy), otherwise\ E(x) = 1\ (alert).$$

Yawning Detection (MAR): This component identifies fatigue by measuring how open someone's mouth is, using something called the Mouth Aspect Ratio. The MAR is calculated like this:

$MAR = (p61 - p69 + p62 - p66 + p63 - p65) / (2 \times p60 - p64$ (3.2.2)

A larger MAR value means the mouth is more open, which usually happens when someone is yawning.

If the MAR goes above a set limit for several frames in a row:

$M(x) = 0$ (meaning yawning is detected), otherwise $M(x) = 1$.

Alcohol Detection: This part checks if the rider has had alcohol by using a gas sensor. The result is shown like this:

$A(x) = 1$ if no alcohol is found

$A(x) = 0$ if alcohol is found

If alcohol is found, the ignition is stopped right away.

Combined Decision Function: All the mentioned components come together to form a single decision model:

$$D(x) = H(x) \times E(x) \times M(x) \times A(x) \quad (3.2.3)$$

If $D(x)$ equals 1, it means all the safety conditions are met and the ignition can proceed.

If $D(x)$ equals 0, it means one or more safety conditions are not met, and the system will stop the ignition or issue an alert.

This organized decision-making process checks several safety factors at the same time. It helps make the system more reliable, reduces mistakes made by humans, and improves rider safety by stopping unsafe situations before they happen.

3.3. System Strategies

Threshold-Based Detection Strategy: The system uses set threshold values to spot unsafe situations. For detecting drowsiness, it checks the Eye Aspect Ratio (EAR) with this formula:

$$EAR = (p2 - p6 + p3 - p5) / (2 \times p1 - p4) \quad (3.3.1)$$

If the EAR is below 0.25 for 20 straight frames, the rider is considered drowsy.

To detect yawning, the system uses the Mouth Aspect Ratio (MAR):

$$MAR = (p61 - p67 + p62 - p66 + p63 - p65) / (2 \times p60 - p64) \quad (3.3.2)$$

If the MAR goes above the MAR_threshold, it means yawning is happening.

These set thresholds help make fast and accurate decisions.

Consecutive Frame Validation: To reduce false detections, the system uses a frame-based validation mechanism:

$$\boldsymbol{Drowsy = 1, if\ (EAR < 0.25)\ for\ N \geq 20\ frames}$$

$$\boldsymbol{Yawning = 1, if\ (MAR > threshold)\ for\ N\ consecutive\ frames}$$

This ensures that temporary eye closure or mouth movement does not trigger incorrect alerts.

Real-Time Processing Strategy: The system processes video frames continuously in real time. For each frame f:

$$\boldsymbol{EAR(f), MAR(f) \rightarrow Decision(f)}$$

This allows the system to monitor rider behavior dynamically and respond immediately when unsafe conditions are detected.

Noise Reduction and Stability: The system combines multiple safety parameters using a decision function:

$$\boldsymbol{D(x) = H(x) \times E(x) \times M(x) \times A(x)}$$

where each parameter represents helmet, eye state, mouth state, and alcohol detection. This ensures that ignition is allowed only when all safety conditions are satisfied.

Wireless Communication Strategy (LoRa): The communication between helmet and vehicle unit is achieved using LoRa. The signal strength and reliability can be represented as:

$$\boldsymbol{Signal} \propto \mathbf{1} / \boldsymbol{Distance}^{n}$$

LoRa provides long-range communication with low power consumption, making it suitable for real-time embedded applications

Chapter 4

RESULTS AND DISCUSSION

4.1. LoRa Wireless Communication (Helmet Unit ↔ Vehicle Unit)

The communication between the helmet unit and the vehicle unit was done using LoRa (Long Range) technology to make sure data is sent reliably over longer distances. The transmitter was built using an Arduino board connected to a LoRa module (RH_RF95 library), while the receiver was made using a Raspberry Pi with Python and the SX127x LoRa library. On the transmitter side, the system starts by setting up the LoRa module with a frequency of 434 MHz and adjusting the power for transmission. A message is created and sent every two seconds. This message shows the status or signal from the helmet. The Arduino code keeps sending the message using the rf95.send() function, allowing the system to deliver data in real time. On the receiver side, the Raspberry Pi is always listening for incoming LoRa signals in receive mode (RXCONT). When a message is received, an interrupt-based function called on_rx_done() is activated. The received data is converted into a readable format and shown along with the RSSI (Received Signal Strength Indicator) value. The RSSI helps understand the quality of the signal and how reliable the communication is. The test results show that the system successfully sends and receives data without major delays or loss. The communication stays stable, and the receiver can process messages in real time. Compared to the earlier wired UART communication used in Phase I, the LoRa-based system offers more flexibility, greater range, and better scalability.

Overall, the LoRa communication module works efficiently and reliably, making it a good choice for real-world use in the safety system where continuous and stable data exchange between the helmet and vehicle is important.

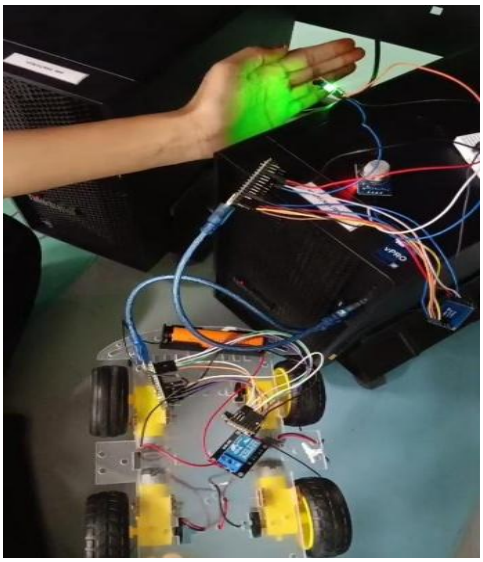


Fig 4(a). Helmet detection using IR sensor detected

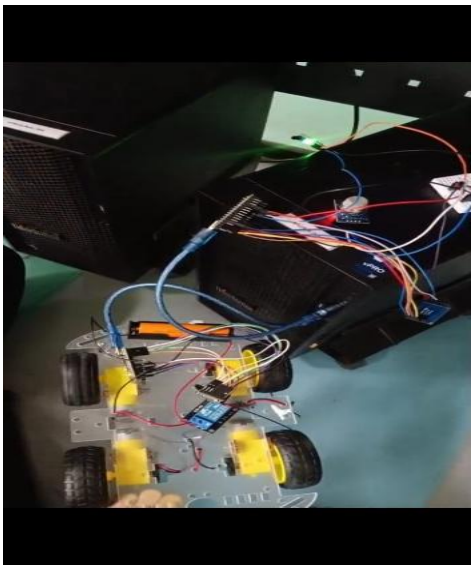


Fig 4(b). Helmet not detected

The figure – 4(a) shows that the helmet detection system works in real time when someone is wearing a helmet. The IR sensor checks whether the rider is present, indicated by a glowing LED. This means the sensor correctly recognizes when the helmet is properly worn. Once the sensor detects the helmet, it sends a signal to the receiver.

This then turns on the ignition system using the relay module. The setup includes a microcontroller, a communication module, and a relay circuit, all working together so the vehicle can only start when the helmet is properly worn.

The result shows the system responds quickly and correctly when the helmet is in place. This proves that the IR sensor works well and ensures the ignition only works when it's safe to start the vehicle.

The figure – 4(b) shows that the helmet detection system works when a helmet is not being worn. In this situation, the IR sensor fails to sense the rider's presence, and the signal it sends shows that no helmet is detected. As a result, the system marks the condition as unsafe. Since the helmet isn't detected, the ignition system stays turned off, and the relay doesn't turn on. This stops the vehicle from starting, making it impossible for the rider to use the bike without a helmet. The hardware setup keeps checking the sensor's output to maintain this safety rule. The outcomes show that the system behaves as intended when the helmet is missing and successfully stops the ignition. This confirms that the detection system is dependable and shows how important it is for ensuring safety in real-time scenarios.

4.2. Drowsiness Detection (EAR-Based System)

The second feature of the system is designed to detect when a rider is getting drowsy, and it uses a vision-based method. This part of the system is built using Python, OpenCV, and Dlib to perform real-time facial landmark detection. The system constantly takes video input from a camera and looks at eye movements to assess how alert the rider is. At the heart of this feature is the Eye Aspect Ratio (EAR), which is calculated by looking at specific points on the face. When the eyes are open, the EAR remains fairly steady, but it drops noticeably when the eyes are closed. By checking this change over time, the system can detect blinking patterns and long periods of eye closure, which are key signs of tiredness. In practice, the system successfully finds facial landmarks and calculates EAR values in real time. To decide if someone is drowsy, a threshold of 0.25 is used. If the EAR stays below this level for 20 consecutive frames, the system labels the rider as drowsy and shows an alert message saying "DROWSY!". If the EAR stays above this threshold, it shows "Awake", meaning the rider is alert. To make the system more reliable, a noise filtering process was added.

This helps eliminate distractions like background noise, changes in lighting, and unwanted movements, so the algorithm can focus only on the face. This leads to more consistent EAR values and fewer false alerts. In addition to tracking eye movement, the system also includes a module that detects yawning by measuring the Mouth Aspect Ratio (MAR). This adds another layer to fatigue detection, making the system more accurate by considering multiple signs of tiredness. The system also handles a special situation where one eye might not be working properly. By identifying the working eye, the system becomes more adaptable and user-friendly.

Overall, the drowsiness detection module works well in real-time, delivering accurate and reliable results. When included in a smart helmet system, it can help identify fatigue early and improve rider safety significantly.

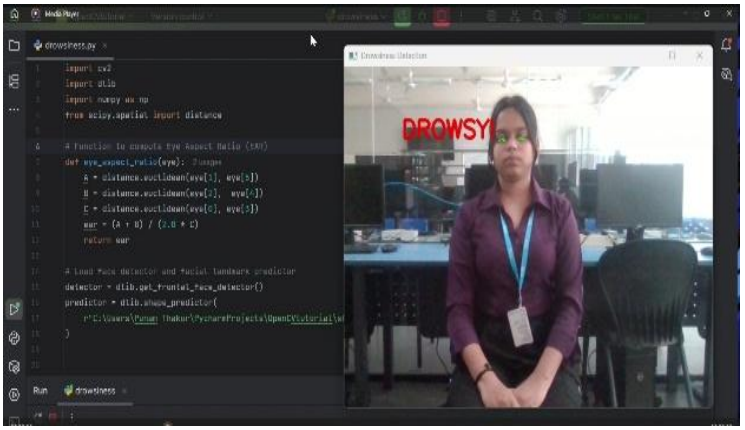


Fig 4(c). Drowsiness Detection (Drowsy)

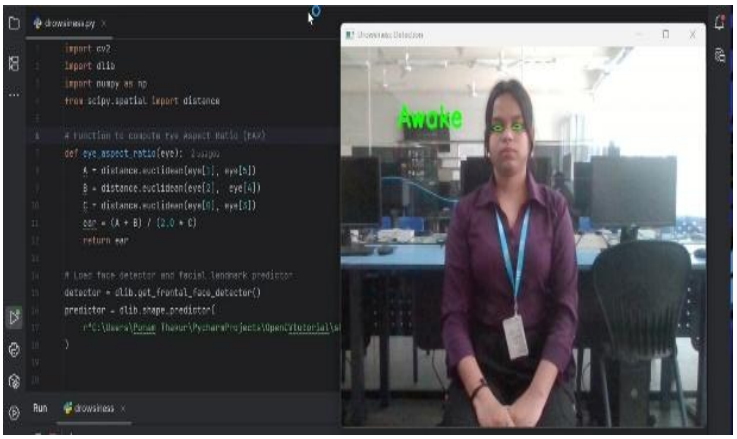


Fig 4(d). Drowsiness Detection (Awake)

The figure – 4(c) shows that system reacts when it detects drowsiness. In this example, the rider's eyes remain closed for a long time, resulting in the Eye Aspect Ratio (EAR) falling below the set threshold of 0.25. If this happens for 20 consecutive frames, the system labels the state as drowsy and shows a "DROWSY!" warning on the screen. The facial landmarks around the eyes confirm that the algorithm is correctly tracking eye movements. This result shows that the system can quickly identify signs of fatigue and issue timely alerts to help prevent accidents.

The figure – 4(d) shows that the system works when the rider is awake. In this case, the eyes are open, and the EAR value is higher than the set limit, which means the eyes are active. Because of this, the screen shows an "Awake" message. The facial features are clearly recognized and stay steady, which means the system is tracking properly. This confirms that the system doesn’t give false warnings when everything is normal and can tell the difference between being alert and feeling sleepy. The system correctly identifies these states, so it only gives alerts when they’re needed, making it more dependable.

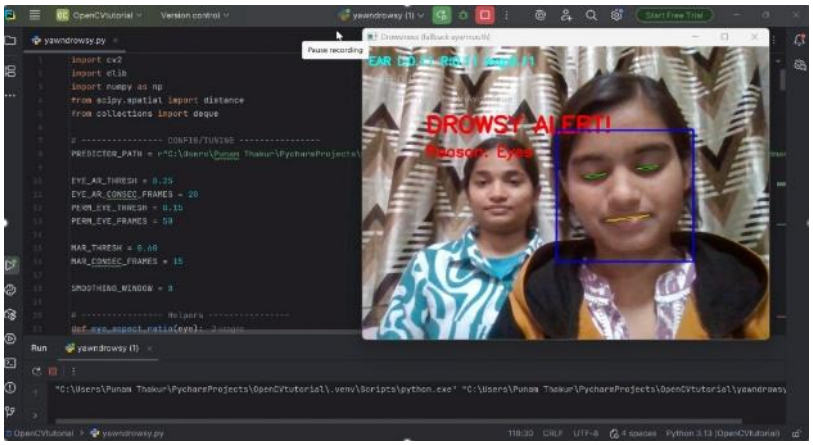


Fig 4(c). Drowsiness Detection (Drowsy)

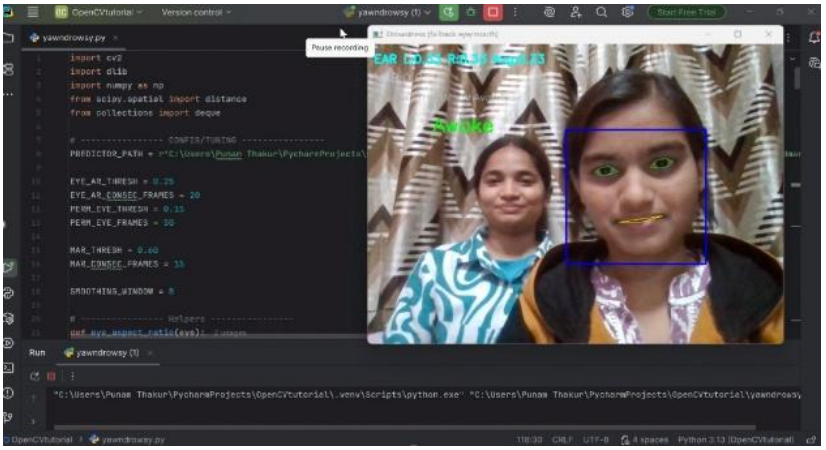


Fig 4(f). Awake State Detection

The figure – 4(e) shows that system identifies when someone is tired by looking at both the Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR). In this example, the rider's eyes are closed, leading to a low EAR value of about 0.11, which is below the set limit. At the same time, the system checks the mouth movements, and together these observations confirm that the person is fatigued. Because of this, the system sends out a "DROWSY ALERT!" and explains that the reason is due to the eyes being closed. The facial features for the eyes and mouth are clearly found and marked, showing the algorithm is working properly. This outcome shows that using both EAR and MAR makes the drowsiness detection more dependable by looking at several signs of tiredness.

The figure – 4(f) shows that the rider being alert. In this case, both eyes are open, and the EAR value is high, around 0.33, which is above the set threshold, showing normal eye movement. The mouth is also in its usual position, and there's no sign of unusual actions like yawning. Because of this, the system shows an "Awake" status, meaning the rider isn't tired. The system consistently detects facial features and keeps the EAR values steady, proving it can clearly tell the difference between being alert and drowsy.

This shows the system doesn't give false warnings and works well for real-time tracking.

4.3. Alcohol Detection

The alcohol detection module is an important safety feature designed to stop drunk driving. It uses an MQ-3 sensor that can detect alcohol levels in the rider's breath. This sensor is connected to an Arduino microcontroller, which constantly reads and checks the sensor data in real time. If the alcohol level goes over a set limit, the system instantly recognizes it as a danger and turns off the vehicle's ignition through a relay. This makes sure the rider can't start the car if they are under the influence of alcohol.

In addition to detecting alcohol, the module is connected to a LoRa wireless system that reliably sends data from the helmet to the vehicle. This wireless connection allows the ignition control to work even without physical wires. The system also works with a Raspberry Pi, which improves its ability to monitor and process information, making it easier to expand and improve in the future. Overall, the alcohol detection module works accurately and quickly, helping to improve rider safety by stopping unsafe driving.

4.4. Comparative analysis:

The system's overall performance is measured by how well it combines several safety features, such as helmet detection, drowsiness detection using EAR and MAR measurements, alcohol detection with an MQ-3 sensor, and wireless communication via LoRa with support from a Raspberry Pi. Each part of the system plays a specific role, and how they work together determines the effectiveness of the smart safety system.

Helmet Detection System:

The helmet detection module works quickly and reliably. The infrared sensor correctly identifies if a helmet is being worn, and the ignition control system reacts right away. The system gives a clear and steady signal, with no noticeable delay when turning the ignition on or off. This module serves as the first line of safety and makes sure basic rules are followed before the vehicle is used.

Drowsiness Detection System (EAR & MAR):

The drowsiness detection module offers precise real-time tracking through computer vision. It uses EAR to detect when someone is closing their eyes for too long, and MAR helps by recognizing yawning, which is another sign of tiredness. The system reacts fast when someone is getting sleepy and sends alerts with very little delay. It also includes noise filtering to make the system more reliable, and partial eye detection makes it work

better in different situations. Even though this module uses more processing power than others, it gives very important safety information.

Alcohol Detection System (MQ-3 Sensor):

The alcohol detection module works well at recognizing high levels of alcohol. The MQ-3 sensor provides reliable results, and the Arduino processes the data precisely to control the ignition system. As soon as alcohol levels go beyond the set limit, the system reacts right away, making sure the car won't start if it's unsafe to do so. This module improves safety by tackling a major real-life danger.

LoRa Communication and System Integration:

LoRa wireless communication ensures reliable data transfer across various parts of the system. This communication remains stable even over long distances with very little delay, making it easy for the helmet, vehicle, and control units to work together smoothly. Adding a Raspberry Pi to the system boosts its functionality by enabling monitoring and data processing.

Comparative Summary:

Among all the modules, the helmet detection system responds the quickest because it uses straightforward logic. The drowsiness detection module, on the other hand, offers the most advanced features, as it continuously monitors and analyzes driver behavior. The alcohol detection system helps enforce safety rules by stopping the vehicle from starting when unsafe conditions are detected. The use of LoRa and Raspberry Pi connects all the modules together, forming a single working system. As a whole, this system is reliable, fast in operation, and effective at enforcing safety, making it ready for real-world use in smart vehicle safety systems.

Chapter 5

CONCLUSION

The project "The Rider's Co-Pilot" offers a comprehensive, smart safety system designed to reduce road accidents caused by human error and unsafe riding conditions. It uses technologies such as embedded systems, computer vision, and wireless communication to ensure critical safety steps are taken automatically before and during vehicle operation. A major success of this project is its use of real-time rider monitoring through vision-based methods. The Eye Aspect Ratio (EAR) algorithm detects drowsiness by analyzing how closed the eyes are, while the Mouth Aspect Ratio (MAR) algorithm flags yawning as a sign of tiredness. These methods, along with checking each video frame, provide accurate and reliable results for detecting unsafe conditions. Plus, the system includes helmet detection to ensure riders are wearing helmets and an alcohol detection feature to stop riding when someone is drunk.

The system also has a robust communication system using LoRa technology, allowing the helmet and vehicle components to reliably share information over long distances. The system uses a rule-based function to evaluate all safety factors simultaneously and control ignition based on the results. This layered design makes the system more dependable and less likely to fail.

In general, the project shows how modern technology can be used together to create a smart and useful safety solution. The system is made to be affordable, easy to expand, and ready for use in real life. By taking care of important safety checks automatically and lessening the need for human decision-making, the system greatly helps improve road safety and encourages better riding habits.

REFERENCES

1. Bhatt, T., Bhatt, S., Nolkha, R. and Ramji, T., 2025, February. Alcohol Detector and Engine Locking System Using Arduino Uno and MQ3 Sensor. In *2025 IEEE International Conference on Emerging Technologies and Applications (MPSec ICETA)* (pp. 1-6). IEEE.

2. Alshaqaqi, B., Baquhaizel, A.S., Ouis, M.E.A., Boumehed, M., Ouamri, A. and Keche, M., 2013, May. Driver drowsiness detection system. In *2013 8th international workshop on systems, signal processing and their applications (WoSSPA)* (pp. 151-155). IEEE.

3. Nataraja, N. and Mamatha, K.S., 2018, May. Smart helmet. In *2018 3rd IEEE International Conference on Recent Trends in Electronics, Information & Communication Technology (RTEICT)* (pp. 2338-2341). IEEE.

4. Chandran, S., Chandrasekar, S. and Elizabeth, N.E., 2016, December. Konnect: An Internet of Things (IoT) based smart helmet for accident detection and notification. In *2016 IEEE Annual India Conference (INDICON)* (pp. 1-4). IEEE.

5. Manjesh, N. and Raj, S., 2014. Smart helmet using GSM & GPS technology for accident detection and reporting system. *International Journal of Electrical and Electronics Research*, *2*(4), pp.122-127.

6. Dharani, P., Ganesh, T., Gopinath, V. and Vali, Y.S., 2020. Smart safety helmet for bike riders using IoT. *International Research Journal of Multidisciplinary Technovation*, *2*(4), pp.21-30.

7. Sobhana, S., Sowmeeya, S.R., Srinathji, M. and Tamilselvan, S., 2021, March. Smart helmet. In *IOP Conference Series: Materials Science and Engineering* (Vol. 1084, No. 1, p. 012116). IOP Publishing.

8. Mathavan, J.J., Wijesekara, V.K.D., Satheeskanth, N., Wanasinghe, W.M.U.J., Maathushan, M. and Wijenayake, V.V., 2022, October. Smart helmet to start the motorbike and to prevent accidents. In *IOP conference series: materials science and engineering* (Vol. 1258, No. 1, p. 012051). IOP Publishing.

9. Titare, S., Chinchghare, S. and Hande, K.N., 2021. Driver drowsiness detection and alert system. *International Journal of Scientific Research in Computer Science, Engineering and Information Technology*, *7*(3), pp.583-588.

10. Saini, V. and Saini, R., 2014. Driver drowsiness detection system and techniques: a review. *International Journal of Computer Science and Information Technologies*, *5*(3), pp.4245-4249.

11. Rode, M., Bethi, S.A., Kondaveti, B.L.P., Gadipally, T. and Katroth, N., 2022. Driver Drowsiness Detection Using Machine Learning. *International Journal for Research in Applied Science & Engineering Technology (IJRASET)*, *10*(11).

12. Rahman, M.A., Ahsanuzzaman, S.M., Rahman, I., Ahmed, T. and Ahsan, A., 2020, June. IoT based smart helmet and accident identification system. In *2020 IEEE Region 10 Symposium (TENSYMP)* (pp. 14-17). IEEE.

13. Shabbeer, S.A. and Meleet, M., 2017, December. Smart helmet for accident detection and notification. In *2017 2nd International Conference on Computational Systems and Information Technology for Sustainable Solution (CSITSS)* (pp. 1-5). IEEE.

14. Pathak, P., 2020, December. Smart Helmet with Motorbike unit for Accident and Rash Driving Detection. In *2020 IEEE International Conference on Advances and Developments in Electrical and Electronics Engineering (ICADEE)* (pp. 1-6). IEEE.

15. Swathi, S.J., Raj, S. and Devaraj, D., 2019, April. Microcontroller and sensor based smart biking system for driver's safety. In *2019 IEEE International Conference on Intelligent Techniques in Control, Optimization and Signal Processing (INCOS)* (pp. 1-5). IEEE.